Endless Line

A handheld Web AR experience about the history of Swiss rail. Scan a ticket card and a looping figure-8 track appears on it. A train runs on it endlessly and changes era as you drag through time, from 1902 to today. No app download is needed.

TYPE	TRACKING	DEVICES	TEAM
Handheld Web AR	**Image target**	**iOS & Android**	**Solo**

A Project idea

The Swiss Federal Railways (SBB) began operating in 1902. Endless Line tells the 124 years since then in a way that fits on a table. A figure-8 (∞) rail track stands for the "endless" network that has kept running for over a century. One train circles it non-stop. With a **timeline slider** the user travels through **five eras**. At each step the train is replaced by the model of that time, from a steam locomotive to today's Giruno. The **sound** (steam, electric hum, modern chime) and the **colour mood** of the scene change with it, with smooth transitions throughout.

Note on scope and branding: this is an unofficial student project. It uses no SBB logos or wordmarks; SBB is only named as historical fact. The goal is a small, finished, polished experience that could later be shown to a rail or museum partner.

1 · Scan
QR on ticket card opens the web app

2 · Start
Loading screen, camera & sound unlock

3 · Track
Ticket recognised, ∞ track appears

4 · Travel
Drag the timeline 1902 → 2026

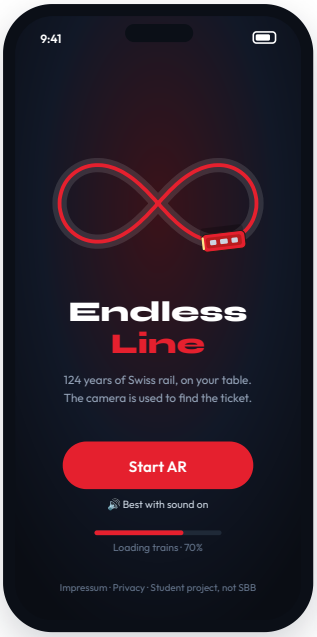
5 · Learn
Tap the train for era facts

B Applied tech stack

AREA	TOOL	WHY
XR framework	A-Frame 1.x + MindAR (image tracking)	Covered in SW02–SW04; open source; Safari + Chrome with one build
Train movement	Custom A-Frame component: parametric ∞ curve (Three.js)	Constant-speed, seamless loop; the train faces along the track
Sound	Web Audio via Howler.js	Crossfades between era sounds; handles the iOS audio unlock
3D assets	Blender (low-poly) → GLB, gltf-transform (Draco + WebP)	5 stylised trains, compressed per SW05
Build & dev	Vite, vanilla JS (ES modules)	HTTPS dev server for phone testing; hashed build files
Version control	GitLab: `main` + `feature/*` branches, merge requests	Clean commit history as required
Hosting & testing	HTTPS static hosting (SW08) · iPhone, Android, desktop	Camera needs HTTPS; weekly real-device tests

A+ Visual concept: the experience on the phone

Everything runs in the phone's browser. The user scans the QR code on the ticket card, points the camera at it, and a looping rail track appears with a train running on it. Dragging the timeline moves through 124 years of Swiss rail: the train, the sound and the colours change smoothly. *Mockups are concept sketches.*



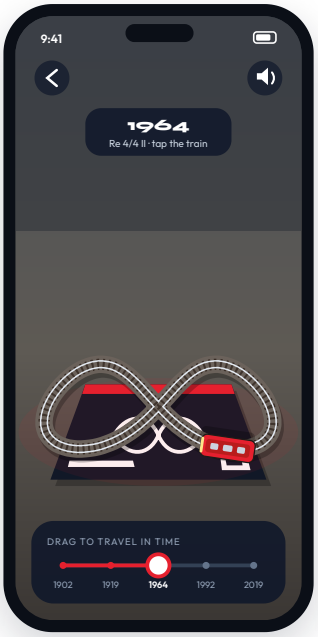
1 Start

Start screen with loading progress. Explains why the camera is used; the tap also unlocks sound.



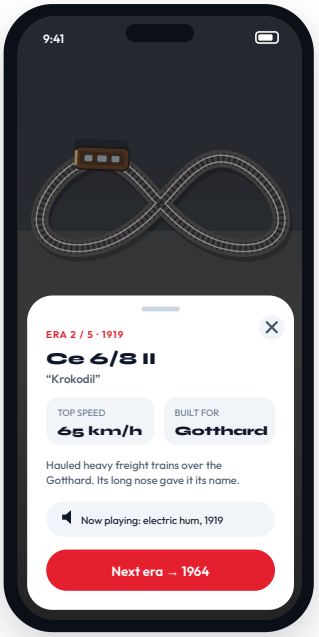
2 Scan

Camera view with a scan frame and hint until the ticket is recognised.



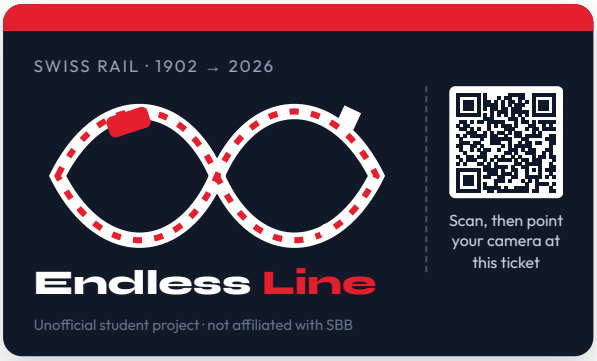
3 Travel in time

The train loops endlessly on the figure-8 track. The timeline slider changes the era.



4 Learn

Tapping the train opens a card with key facts and the era's sound.



The ticket card (A6): the AR target

A ticket-shaped card is both the giveaway and the image target. Its graphic is built to track well: **high contrast, sharp edges, asymmetric details** (SW04). No SBB logo or wordmark is used.

Interactions

Drag slider	Travel 1902 → 2026; train, sound and colours change
Tap train	Info card with key facts for that era
Tap 🔊	Sound on / off
Move phone	View the track from any angle
Lose ticket	Scene hides, hint returns
No camera	Same scene in the browser, drag to orbit

c Feature specification & acceptance criteria

The features follow the user's path through the app. SW06 usability measures are built into the features they belong to (overview on page 5).

Feature 1 · First impression & loading screen

CORE

Something intentional is visible right away, the shared link looks good, and loading never shows a blank page.

- Custom `<title>`, favicon and Open Graph tags (og:title, og:image)
- Shared link shows preview image + title in WhatsApp/Teams
- Logo + title visible ≤ 1 s after opening (no blank screen)
- Progress bar with text ("Loading trains… 70 %") from `a-asset-item` progress
- Loader hides the page until CSS and fonts are ready (no FOUC / FOUT)
- Loader disappears when `<a-scene>` fires `loaded`

Feature 2 · Start, permissions, errors & fallback

CORE

The app checks what the device supports, explains the camera, unlocks sound with the start tap, and never ends in a dead end.

- Feature detection on load: HTTPS, camera (getUserMedia), WebGL
- "Start AR" only shown if checks pass; otherwise the 3D fallback is offered
- Start screen says why the camera is needed and that sound is part of it
- Camera and audio only start after the tap on "Start AR"
- Camera denied → human message + how to re-enable it in browser settings
- No internet / model fails → message + "Reload" button
- Fallback: same scene in the browser, drag to orbit (touch + mouse)
- Failure paths tested: denied permission, airplane mode, broken file name

Feature 3 · Track the ticket & the endless loop

CORE

The ticket card is recognised and a figure-8 track is anchored on it. A train runs along it continuously and smoothly.

- Ticket recognised within 2 s; scene stays centred on it when moved
- Tracking works at 30–60 cm distance
- Train follows the ∞ curve at constant speed and always faces along the track
- Loop is seamless: no jump or stutter at the crossing / loop start
- Runs at ≥ 30 FPS on a mid-range phone (tested on 2 devices)
- "Point your camera at the ticket" hint until found, hidden when found
- Tracking lost > 1 s → scene hides + "Ticket lost, point at it again"
- MindAR default scanning/error UI replaced with own styled elements

Feature 4 · Timeline: travel through five eras

CORE

A slider lets the user move from 1902 to today. The train, colours and year label change with smooth transitions.

- 5 eras: 1902 steam (A 3/5) · 1919 Ce 6/8 II "Krokodil" · 1964 Re 4/4 II · 1992 Re 460 · 2019 Giruno*
- Slider snaps to the 5 eras; usable with one thumb, handle ≥ 44 px
- Train swap via fade/scale transition ≤ 600 ms, the train keeps moving during it
- Year label counts up/down to the new year during the transition
- Scene colour mood changes per era (sepia → full colour)
- Optional "Play" button auto-runs through all eras (≈ 5 s per era)
- Fast slider dragging never shows two trains or a missing train

Feature 5 · Sound

CORE

Each era has its own sound, so the history can be heard as well as seen. Sound is optional and never surprises the user.

- One ambient loop per era (steam, early electric, modern), CC0 / own recordings
- Era change → crossfade between loops ≤ 1 s, plus a short whistle / chime
- Audio starts only after the "Start AR" tap (works on iOS Safari)
- Mute button always visible; the choice is kept for the session
- Sound pauses when tracking is lost or the tab is hidden
- All audio files compressed (≤ 300 KB each, AAC/MP3)

Feature 6 · Train info, end state & production polish

CORE

Tapping the train shows key facts. The app ends with a clear next step and meets basic production standards.

- Tap train → info card: name, year, top speed, one-sentence fact
- Cards are an HTML overlay; × or tap outside closes; text ≥ 16 px, targets ≥ 44 px
- After the last era: "Ride again" + link to a source (e.g. SBB Historic)
- Impressum · Privacy · "not affiliated with SBB" link on every screen
- Privacy note: camera images stay on the device; no analytics, no cookies
- Viewport meta set; no accidental scroll/zoom; Wake Lock keeps screen on
- Train GLBs Draco-compressed, ≤ 3 MB total; first era preloaded, others deferred
- One palette + font pair as CSS variables; same button style everywhere; hosted on HTTPS

Long shots · if time allows

STRETCH

- **Scenery per era:** small low-poly landscape pieces (tunnel, Alps, station) inside the loop that change with the era.
- **Languages:** DE / FR / IT / EN for all texts, switchable without reload.
- **Gestures:** pinch to scale and swipe to rotate the whole track (0.5×–2×).

* Exact years and facts are verified during the research step (SBB Historic, Wikipedia) before the content is final.

D How I plan to use AI

- **Coding assistant (Claude):** one feature at a time, with that feature's acceptance criteria from this proposal as the definition of "done".
- **Transparency:** commits with AI-generated code are tagged `[AI]` in the message, with a note on what was generated.
- **Debugging & research help:** explaining errors, the ∞ curve maths, first drafts of era facts that I check against sources.
- **No one-shotting:** every AI suggestion is reviewed and tested on a real device before merging.
- **Not AI:** concept, ticket design, train models, sound choice and final texts are done by me.

E Usability checklist (SW06) → where it is covered

SW06 CHECKLIST ITEM	COVERED IN	HOW
Something intentional visible right after opening	F1	Logo + title within 1 s
Loading shows progress or text	F1	"Loading trains… %" bar
Start button explains why the camera is needed	F2	One sentence on the start screen
Hint on how to start (what to point at)	F3	"Point your camera at the ticket" + scan frame
Message when tracking is lost	F3	"Ticket lost…" after 1 s, sound pauses
Clear message when camera denied / not supported	F2	Feature detection + human error messages
Fallback so there is no dead end	F2	Same scene in the browser, orbit by touch / mouse
Big model loaded only when needed	F6	First era preloaded, other trains deferred
Text and buttons readable on the real device	F4 · F6	≥ 16 px text, ≥ 44 px targets, HTML overlay
Title, icon, share preview	F1	`<title>`, favicon, og:title / og:image
One look everywhere	F3 · F6	CSS variables, one button style, restyled MindAR UI
Tested on real devices, bad network, permissions denied	F2	iPhone + Android, airplane mode, denied permission

F Milestones

30.10	Revised proposal sent · GitLab repo & Vite setup
04.11	F1 loader · F2 start, errors, fallback
18.11	F3 ticket tracking + ∞ loop with 1 train
25.11	F4 timeline + all 5 trains
02.12	F5 sound · F6 info cards & polish
09.12	Hosting, failure-path & device testing
16.12	Demo station & oral report

G Risks & mitigation

5 train models take too long	Stylised low-poly (shape + colour), one shared base mesh; CC-licensed models as backup
Audio on iOS	Unlock on the start tap; test on iPhone from F5 day one
Branding / trademark	No SBB logos, marked as unofficial on every screen
Running out of time	Long shots only after F1–F6; F5 can drop to one sound per era